

# Introduction to Electronics Engineering

## MODULE 4 — PART 2: QUESTION BANK

### Module 4 (07 hrs): Introduction to Digital Electronics

For First-Year Students — with complete answers and diagrams  
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

#### Question Paper Structure

| Section   | Marks per Question | Scope                          | Questions |
|-----------|--------------------|--------------------------------|-----------|
| Section A | 2 marks            | Compulsory — from every module | Q1 – Q28  |
| Section B | 7 marks            | Compulsory — from every module | Q1 – Q12  |

- Module 4 has **no 5-mark section** — 5-mark questions are asked only from Modules 1 and 2.
- Every topic of Module 4 has a 2-mark question, and every topic is also covered in the 7-mark section — no topic is left out.
- No question is repeated between sections.
- Answer 2-mark questions in 2–4 lines and 7-mark questions with full explanation, diagrams and truth tables.

## SECTION A — 2-Mark Questions

### Q1. What is Boolean algebra?

**Answer:** Boolean algebra is a branch of mathematics dealing with variables that have only two values: True (1) or False (0). It was introduced by George Boole in 1854 and is the mathematical basis of digital circuits.

### Q2. What is a logic gate? How many types of gates are there?

**Answer:** A logic gate is a device that performs a basic Boolean operation on one or more binary inputs to produce a single binary output. There are seven types: NOT, AND, OR, XOR, XNOR, NAND and NOR.

### Q3. What are the three basic Boolean operations and their symbols?

**Answer:** AND (written  $x \cdot y$  or  $xy$ ), OR (written  $x + y$ ) and NOT (written  $x'$ , the complement).

### Q4. What are logic 0 and logic 1?

**Answer:** Logic 0 is a low voltage signal (around 0 volts) and logic 1 is a high voltage signal (e.g. 5 V or 3.3 V). The physical value of a signal is the actual voltage, while its logic value is either 0 (low) or 1 (high).

### Q5. In how many ways can the behaviour of a gate be described?

**Answer:** In three ways: Boolean expressions, logic diagrams, and truth tables.

### Q6. Write the Boolean expression and truth table of the NOT gate.

**Answer:** A NOT gate (inverter) has one input and one output:  $Y = A'$ . If the input is 0 the output is 1, and if the input is 1 the output is 0.

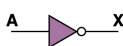
| Boolean Expression | Logic Diagram Symbol                                                                | Truth Table                                                                                            |   |   |   |   |   |   |
|--------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
| $X = A'$           |  | <table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table> | A | X | 0 | 1 | 1 | 0 |
| A                  | X                                                                                   |                                                                                                        |   |   |   |   |   |   |
| 0                  | 1                                                                                   |                                                                                                        |   |   |   |   |   |   |
| 1                  | 0                                                                                   |                                                                                                        |   |   |   |   |   |   |

Figure 1: NOT gate — symbol and truth table.

### Q7. Write the Boolean expression of the AND gate and state when its output is 1.

**Answer:**  $Y = A \cdot B$ . The output is 1 only when **both** inputs are 1; otherwise the output is 0.

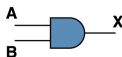
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| A                  | B                                                                                   | X                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                   | 0                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                   | 0                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                   | 0                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                   | 1                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 2: AND gate — symbol and truth table.

**Q8. Write the Boolean expression of the OR gate and state when its output is 0.**

**Answer:**  $Y = A + B$ . The output is 0 only when **both** inputs are 0; otherwise the output is 1.

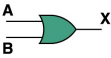
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| A                  | B                                                                                 | X                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                 | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 3: OR gate — symbol and truth table.

**Q9. What is an XOR gate? How does it differ from an OR gate?**

**Answer:** An XOR (exclusive OR) gate produces 0 when its two inputs are the same and 1 otherwise:  $Y = A'B + AB'$ . It differs from the OR gate only when both inputs are 1 — the OR gives 1 but the XOR gives 0.

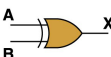
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| A                  | B                                                                                 | X                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                 | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                 | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 4: XOR gate — symbol and truth table.

**Q10. What is an XNOR gate?**

**Answer:** The XNOR gate is the complement of the XOR gate. It produces 1 when both inputs are the same and 0 otherwise:  $Y = (A \oplus B)' = AB + A'B'$ .

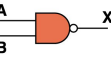
| Boolean Expression  | Logic Diagram Symbol                                                                | Truth Table                                                                                                                                                                                                                                                            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
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| A                   | B                                                                                   | X                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                   | 0                                                                                   | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                   | 1                                                                                   | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                   | 0                                                                                   | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                   | 1                                                                                   | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 5: XNOR gate — symbol and truth table.

**Q11. What is a NAND gate?**

**Answer:** The NAND gate is the opposite of the AND gate:  $Y = (A \cdot B)'$ . Its output is 0 only when both inputs are 1.

**Q12. What is a NOR gate?**

**Answer:** The NOR gate is the opposite of the OR gate:  $Y = (A + B)'$ . Its output is 1 only when both inputs are 0.

**Q13. What is meant by a three-input AND gate?**

**Answer:** A gate designed to accept three or more inputs. A three-input AND gate produces an output of 1 only if **all** its input values are 1.

**Q14. State any four laws of Boolean algebra.**

**Answer:** Identity:  $x + 0 = x$ ,  $x \cdot 1 = x$ . Complement:  $x + x' = 1$ ,  $x \cdot x' = 0$ . Idempotence:  $x + x = x$ ,  $x \cdot x = x$ . Distributive:  $x(y + z) = xy + xz$ .

**Q15. State De Morgan's theorems.**

**Answer:** First law:  $(A + B)' = A' \cdot B'$  (the complement of a sum equals the product of the complements). Second law:  $(A \cdot B)' = A' + B'$  (the complement of a product equals the sum of the complements).

**Q16. Why are NAND and NOR called universal gates?**

**Answer:** Because they are self-sufficient — any logic circuit can be built using only NAND gates, or only NOR gates. They can replace an AND gate, an OR gate or an inverter.

**Q17. How do you make a NOT gate using a NAND gate?**

**Answer:** Tie both inputs of the NAND gate together:  $Z = (X \cdot X)' = X'$ . The circuit is equivalent to an inverter.

**Q18. How do you make an AND gate using NAND gates?**

**Answer:** Use a NAND gate followed by an inverter (a second NAND with its inputs tied together):  $Z = ((X \cdot Y)')' = X \cdot Y$ . Two NAND gates are needed.

**Q19. How do you make an OR gate using NAND gates?**

**Answer:** Invert both inputs and then apply NAND:  $Z = X' \cdot Y' = (X + Y)'$ . Three NAND gates are needed.

**Q20. How do you make a NOT gate using a NOR gate?**

**Answer:** Tie both inputs of the NOR gate together:  $Z = (X + X)' = X'$ . The circuit acts as an inverter.

**Q21. What is the difference between a combinational circuit and a sequential circuit?**

**Answer:** In a combinational circuit the output depends only on the present input values. In a sequential circuit the output depends on the input values as well as the existing state of the circuit.

**Q22. What is a half adder? Write its Boolean expressions.**

**Answer:** A half adder is a circuit that computes the sum of two bits and produces the correct carry bit.  $\text{Sum} = A \oplus B$  and  $\text{Carry} = A \cdot B$ .

**Q23. What is a full adder? Write its Boolean expressions.**

**Answer:** A full adder is a combinational circuit that forms the arithmetic sum of three input bits. It has three inputs ( $A$ ,  $B$ ,  $C_{in}$ ) and two outputs.  $\text{Sum} = A \oplus B \oplus C_{in}$  and  $\text{Carry-out} = AB + C_{in}(A \oplus B)$ .

**Q24. What is a half subtractor? Write its Boolean expressions.**

**Answer:** A half subtractor performs the subtraction of two single-bit binary numbers and gives two outputs — difference ( $D$ ) and borrow ( $B$ ).  $D = A \oplus B$  and  $\text{Borrow } B = A' \cdot B$ .

**Q25. Define SOP and POS.**

**Answer:** SOP (Sum of Products) is a set of product (AND) terms summed (OR) together — it uses minterms. POS (Product of Sums) is a set of sum (OR) terms multiplied (AND) together — it uses maxterms.

**Q26. What is the difference between standard form and canonical form?**

**Answer:** In standard form, terms may have missing literals (e.g.  $F = AB + AC$ ). In canonical form every term contains ALL the variables exactly once (e.g.  $F = A'BC + AB'C + ABC' + ABC$ ).

**Q27. Define minterm and maxterm.**

**Answer:** A minterm is a product term that contains all the variables exactly once (used in canonical SOP); it is denoted by 'm'. A maxterm is a sum term that contains all the variables exactly once (used in canonical POS); it is denoted by 'M'.

**Q28. How do you complement a Boolean function?**

**Answer:** By interchanging the AND and OR operators and complementing every literal (using De Morgan's theorems). For example, if  $f = x'yz' + xy'z'$ , then  $f' = (x + y' + z)(x' + y + z)$ .

**SECTION B — 7-Mark Questions****Q1. Explain Boolean algebra and its importance in digital electronics. What are logic 0 and logic 1?**

**Answer:**

**Boolean algebra** is a branch of mathematics dealing with variables that have only two values: True (1) or False (0). It was introduced by George Boole in 1854. Its three basic operations are AND ( $x \cdot y$ ), OR ( $x + y$ ) and NOT ( $x'$ ).

**Importance:** a logic gate performs a Boolean function on binary inputs to give a single binary output. Digital circuits use two-valued signals — logic 0 is a low voltage (about 0 V) and logic 1 is a high voltage (5 V or 3.3 V). The physical value of a signal is the actual voltage it carries, while its logic value is 0 or 1. Having only two values greatly simplifies the implementation of digital circuits.

- Introduced by George Boole in 1854
- Three basic operations: **AND**, **OR**, and **NOT**
- The **AND** operator is denoted by a dot ( $\cdot$ ),  
 $x \cdot y$  or  $xy$  is read:  $x$  **AND**  $y$
- The **OR** operator is denoted by a plus (+)  
 $x + y$  is read:  $x$  **OR**  $y$
- The **NOT** operator is denoted by (') or an overbar ( $\bar{\phantom{x}}$ ).  
 $x'$  or  $\bar{x}$  is the complement of  $x$

$x \cdot y$  is the **AND** operator  
 $x + y$  is the **OR** operator  
 $x'$  is the **NOT** operator

Figure 6: The three basic Boolean operations.

| AND (.)                                       | OR (+)                                         | NOT (—)                                |
|-----------------------------------------------|------------------------------------------------|----------------------------------------|
| $0 \cdot 0 = 0$                               | $0 + 0 = 0$                                    | $\bar{0} = 1$                          |
| $0 \cdot 1 = 0$                               | $0 + 1 = 1$                                    | $\bar{1} = 0$                          |
| $1 \cdot 0 = 0$                               | $1 + 0 = 1$                                    |                                        |
| $1 \cdot 1 = 1$                               | $1 + 1 = 1$                                    |                                        |
| Multiplication and AND give identical results | Addition and OR give different Results for 1+1 | No corresponding Math operator for NOT |

Figure 7: Truth tables of AND, OR and NOT.

**Q2. Explain the basic logic gates NOT, AND and OR with their symbols, Boolean expressions and truth tables.**

**Answer:**

**NOT gate:** one input, one output; it inverts the input,  $Y = A'$ . If  $A = 0$ ,  $Y = 1$ ; if  $A = 1$ ,  $Y = 0$ . It is also called an inverter.

**AND gate:** two inputs;  $Y = A \cdot B$ . The output is 1 only when both inputs are 1.

**OR gate:** two inputs;  $Y = A + B$ . The output is 0 only when both inputs are 0.

Gates can also be built with three or more inputs — for example, a three-input AND gate gives 1 only when all three inputs are 1.

| Boolean Expression | Logic Diagram Symbol                                                                | Truth Table                                                                                            |   |   |   |   |   |   |
|--------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
| $X = A'$           |  | <table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table> | A | X | 0 | 1 | 1 | 0 |
| A                  | X                                                                                   |                                                                                                        |   |   |   |   |   |   |
| 0                  | 1                                                                                   |                                                                                                        |   |   |   |   |   |   |
| 1                  | 0                                                                                   |                                                                                                        |   |   |   |   |   |   |

Figure 8: NOT gate.

| Boolean Expression | Logic Diagram Symbol                                                                | Truth Table                                                                                                                                                                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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| A                  | B                                                                                   | X                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                   | 0                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                   | 0                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                   | 0                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                   | 1                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 9: AND gate.

| Boolean Expression | Logic Diagram Symbol                                                                | Truth Table                                                                                                                                                                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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| A                  | B                                                                                   | X                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                   | 0                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                   | 1                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                   | 1                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                   | 1                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 10: OR gate.

### Q3. Explain the XOR, XNOR, NAND and NOR gates with symbols and truth tables. How does XOR differ from OR?

**Answer:**

**XOR gate:**  $Y = A'B + AB'$ ; the output is 0 when the inputs are the same and 1 otherwise. It differs from OR only when both inputs are 1 (OR gives 1, XOR gives 0).

**XNOR gate:** the complement of XOR,  $Y = AB + A'B'$ ; the output is 1 when the inputs are the same.

**NAND gate:** the opposite of AND,  $Y = (A \cdot B)'$ ; the output is 0 only when both inputs are 1.

**NOR gate:** the opposite of OR,  $Y = (A + B)'$ ; the output is 1 only when both inputs are 0.

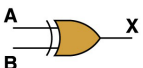
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| A                  | B                                                                                 | X                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                 | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                 | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                 | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 11: XOR gate.

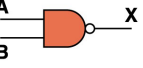
| Boolean Expression | Logic Diagram Symbol                                                               | Truth Table                                                                                                                                                                                                                                                            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $X = (A \cdot B)'$ |  | <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>X</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> | A | B | X | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| A                  | B                                                                                  | X                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 0                                                                                  | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                  | 1                                                                                  | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 0                                                                                  | 1                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                  | 1                                                                                  | 0                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 12: XNOR gate.

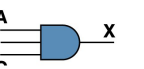
| Boolean Expression      | Logic Diagram Symbol                                                                | Truth Table                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $X = A \cdot B \cdot C$ |  | <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr></table> | A | B | C | X | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| A                       | B                                                                                   | C                                                                                                                                                                                                                                                                    | X |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                       | 0                                                                                   | 0                                                                                                                                                                                                                                                                    | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                       | 0                                                                                   | 1                                                                                                                                                                                                                                                                    | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                       | 1                                                                                   | 0                                                                                                                                                                                                                                                                    | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                       | 1                                                                                   | 1                                                                                                                                                                                                                                                                    | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 13: NAND and NOR gates.

### Q4. Explain the properties (laws) of Boolean algebra.

**Answer:**

The main laws of Boolean algebra are:

| Property     | Form 1              | Dual Form                                   |
|--------------|---------------------|---------------------------------------------|
| Identity     | $x + 0 = x$         | $x \cdot 1 = x$                             |
| Complement   | $x + x' = 1$        | $x \cdot x' = 0$                            |
| Null         | $x + 1 = 1$         | $x \cdot 0 = 0$                             |
| Idempotence  | $x + x = x$         | $x \cdot x = x$                             |
| Involution   | $(x')' = x$         | —                                           |
| Commutative  | $x + y = y + x$     | $x \cdot y = y \cdot x$                     |
| Associative  | $(x+y)+z = x+(y+z)$ | $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ |
| Distributive | $x(y+z) = xy + xz$  | $x + yz = (x+y)(x+z)$                       |
| Absorption   | $x + xy = x$        | $x(x+y) = x$                                |

| Property       | Form 1            | Dual Form         |
|----------------|-------------------|-------------------|
| Simplification | $x + x'y = x + y$ | $x(x'+y) = xy$    |
| De Morgan      | $(x+y)' = x'y'$   | $(xy)' = x' + y'$ |

1. Closure: the result of any Boolean operation is in  $B = \{0, 1\}$

2. Identity element with respect to + is 0:  $x + 0 = 0 + x = x$

Identity element with respect to  $\cdot$  is 1:  $x \cdot 1 = 1 \cdot x = x$

3. Commutative with respect to +:  $x + y = y + x$

Commutative with respect to  $\cdot$ :  $x \cdot y = y \cdot x$

4.  $\cdot$  is distributive over +:  $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$

Figure 14: Basic Boolean-algebra properties.

### Q5. State and prove De Morgan's laws with gate realizations and truth tables.

**Answer:**

**De Morgan's theorems** allow an AND of complemented terms to be converted into an OR of uncomplemented terms and vice versa:

**First law:**  $(A + B)' = A' \cdot B'$       **Second law:**  $(A \cdot B)' = A' + B'$

The first law says the complement of a sum equals the product of the complements; the second says the complement of a product equals the sum of the complements. The laws are verified by the truth tables shown below and are widely used to simplify Boolean expressions.

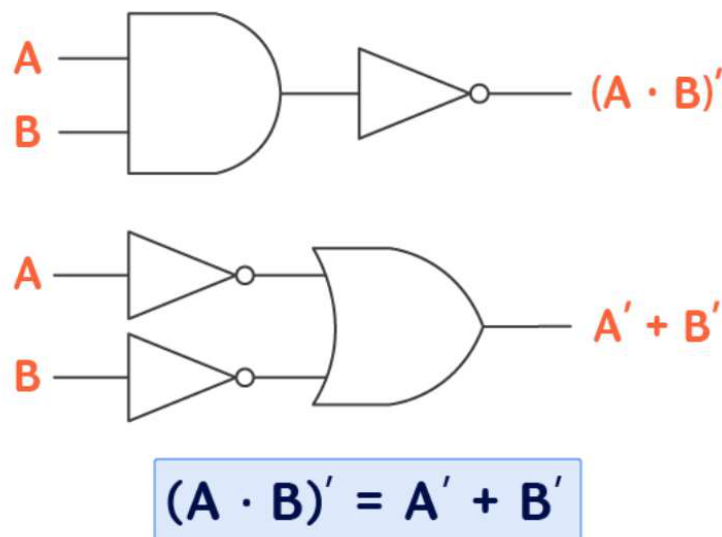


Figure 15: De Morgan's laws realized with gates.

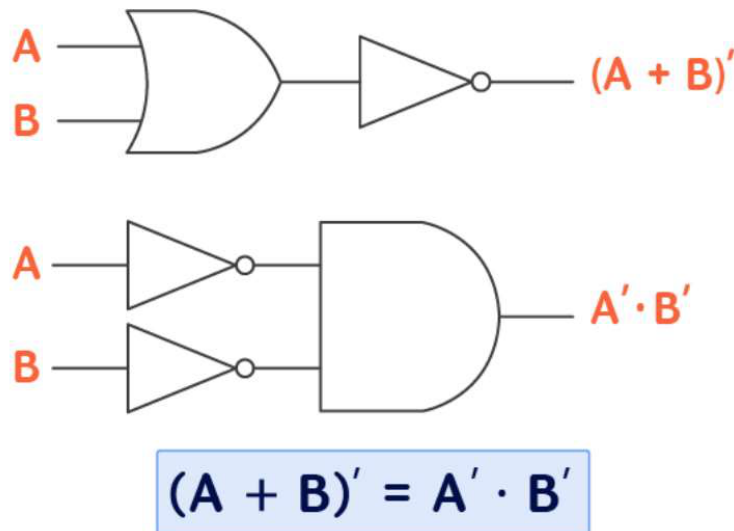


Figure 16: Truth table for the first law.

| Inputs |   |             | Outputs        |      |      |           |
|--------|---|-------------|----------------|------|------|-----------|
| A      | B | $A \cdot B$ | $(A \cdot B)'$ | $A'$ | $B'$ | $A' + B'$ |
| 0      | 0 | 0           | 1              | 1    | 1    | 1         |
| 0      | 1 | 0           | 1              | 1    | 0    | 1         |
| 1      | 0 | 0           | 1              | 0    | 1    | 1         |
| 1      | 1 | 1           | 0              | 0    | 0    | 0         |

Figure 17: Truth table for the second law.

**Q6. Explain how the basic gates (NOT, AND, OR) are realized using NAND gates. Why is NAND a universal gate?**

**Answer:**

NAND is a **universal gate**: any logic circuit can be built using only NAND gates, so NAND forms a complete set of logic.

- **NOT using NAND:** tie both inputs together —  $Z = (X \cdot X)' = X'$ .
- **AND using NAND:** a NAND gate followed by an inverter —  $Z = ((X \cdot Y)')' = X \cdot Y$ .
- **OR using NAND:** invert both inputs and apply NAND —  $Z = X' \cdot Y' = (X + Y)'$ .

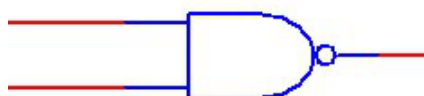


Figure 18: NOT using NAND.

■ Implementing AND using NAND gates:

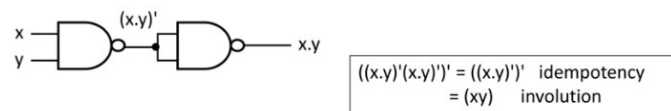


Figure 19: AND using NAND gates.

■ Implementing OR using NAND gates:

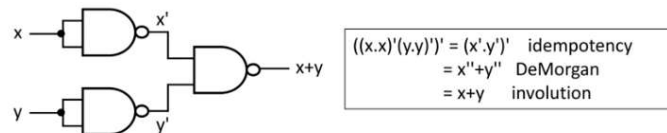


Figure 20: OR using NAND gates.

**Q7. Explain how the basic gates (NOT, OR, AND) are realized using NOR gates. Why is NOR a universal gate?**

**Answer:**

NOR is also a **universal gate**: any logic circuit can be built using only NOR gates.

- **NOT using NOR**: tie both inputs together —  $Z = (X + X)' = X'$ .
- **OR using NOR**: a NOR gate followed by an inverter —  $Z = ((X + Y)')' = X + Y$ .
- **AND using NOR**: invert both inputs and apply NOR —  $Z = X' + Y' = (X \cdot Y)'$ .

• NOR Gate as an Inverter



Figure 21: NOT using NOR.

■ Implementing OR using NOR gates:

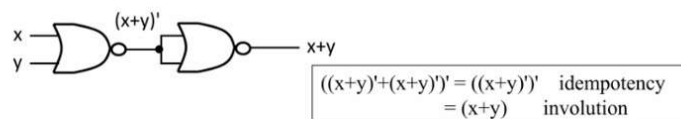


Figure 22: OR using NOR gates.

■ Implementing AND using NOR gates:

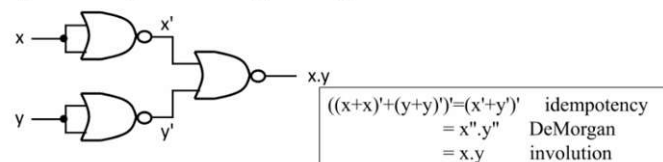


Figure 23: AND using NOR gates.

**Q8. Explain combinational and sequential circuits. Give an example of a combinational circuit and draw its truth table.**

**Answer:**

Digital circuits fall into two general categories. In a **combinational circuit** the input values explicitly determine the output; in a **sequential circuit** the output is a function of the input values as well as the existing state of the circuit. The operation of a circuit can be described by Boolean expressions, logic diagrams or truth tables.

**Example:** gates are combined by using the output of one gate as the input of another. The circuit for the function  $(AB + AC)$  has three inputs, so eight rows are required in its truth table.

| A | B | C | D | E | X |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 |

Figure 24: Combinational circuit for  $AB + AC$  with its truth table.

**Q9. Explain the half adder and the full adder with circuits, Boolean expressions and truth tables.**

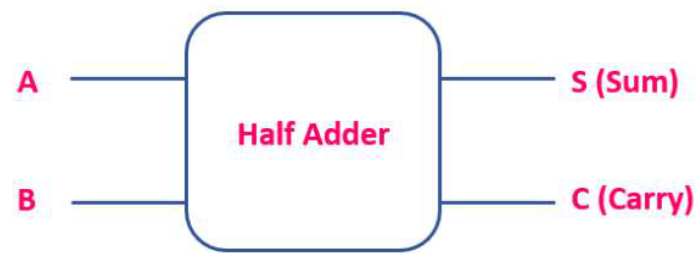
**Answer:**

**Half adder:** adds two binary bits and produces a sum and a carry.  $\text{Sum} = A \oplus B$  and  $\text{Carry} = A \cdot B$ . It is built from an XOR gate (sum) and an AND gate (carry).

**Full adder:** a combinational circuit that forms the arithmetic sum of three input bits; it has three inputs ( $A, B, C_{in}$ ) and two outputs (Sum, Carry-out).

$$\text{Sum} = A \oplus B \oplus C_{in} \quad \text{Carry-out} = A \cdot B + C_{in}(A \oplus B)$$

A full adder can be built from two half adders and an OR gate.



| A | B | Sum | Carry |
|---|---|-----|-------|
| 0 | 0 | 0   | 0     |
| 0 | 1 | 1   | 0     |
| 1 | 0 | 1   | 0     |
| 1 | 1 | 0   | 1     |

Figure 25: Half adder block diagram and truth table.

$$S = \overline{A}B + A\overline{B} = A \oplus B$$

$$C = AB$$

### Logic Circuit

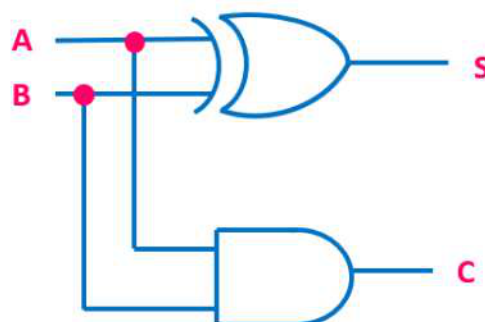


Figure 26: Half adder logic circuit.

| A | B | C <sub>in</sub> | Sum | C <sub>out</sub> |
|---|---|-----------------|-----|------------------|
| 0 | 0 | 0               | 0   | 0                |
| 0 | 0 | 1               | 1   | 0                |
| 0 | 1 | 0               | 1   | 0                |
| 0 | 1 | 1               | 0   | 1                |
| 1 | 0 | 0               | 1   | 0                |
| 1 | 0 | 1               | 0   | 1                |
| 1 | 1 | 0               | 0   | 1                |
| 1 | 1 | 1               | 1   | 1                |

Figure 27: Full adder truth table.

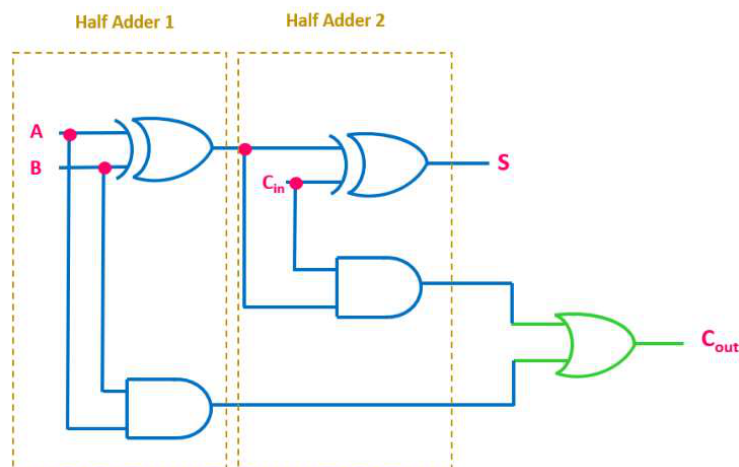


Figure 28: Full adder using two half adders and an OR gate.

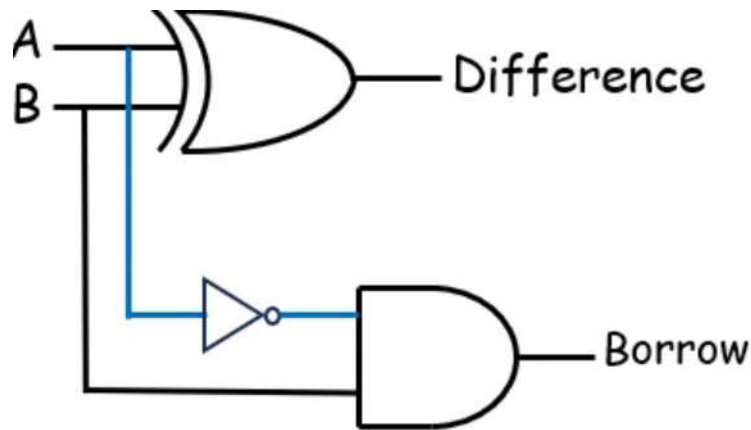
**Q10. Explain the half subtractor with its circuit diagram, Boolean expressions and truth table.**

**Answer:**

A **half subtractor** is a digital circuit that performs the subtraction of two single-bit binary numbers. It produces two outputs: a difference bit (D) and a borrow bit (B).

$$\text{Difference } D = A \oplus B \quad \text{Borrow } B = A' \cdot B$$

It is built from an XOR gate (which computes the difference) and an AND gate with an inverted A (which computes the borrow).



Half Subtractor Circuit Diagram

Figure 29: Half subtractor circuit.

| A | B | Difference | Borrow |
|---|---|------------|--------|
| 0 | 0 | 0          | 0      |
| 0 | 1 | 1          | 1      |
| 1 | 0 | 1          | 0      |
| 1 | 1 | 0          | 0      |

Figure 30: Half subtractor truth table.

**Q11. Explain SOP and POS forms. What is the difference between standard and canonical forms? Explain minterms and maxterms.**

**Answer:**

**SOP (Sum of Products):** a set of product (AND) terms summed (OR) together; it uses minterms and gives a HIGH (1) output. **POS (Product of Sums):** a set of sum (OR) terms multiplied (AND) together; it uses maxterms and gives a LOW (0) output.

| S.No. | SOP                                     | POS                                          |
|-------|-----------------------------------------|----------------------------------------------|
| 1     | Sum of product terms                    | Product of sum terms                         |
| 2     | Prefers minterms ('m')                  | Prefers maxterms ('M')                       |
| 3     | Gives a HIGH (1) output                 | Gives a LOW (0) output                       |
| 4     | Terms added to get the final expression | Terms multiplied to get the final expression |

**Standard vs canonical:** in standard form the terms may have missing literals (e.g.  $F = AB + AC$ ); in canonical form every term contains ALL the variables exactly once (e.g.  $F = A'BC + AB'C + ABC' + ABC$ ). Minterms give the canonical SOP form and maxterms give the canonical POS form.

| Variables |   |   | Minterms                        | Maxterms                            |
|-----------|---|---|---------------------------------|-------------------------------------|
| A         | B | C | $m_i$                           | $M_i$                               |
| 0         | 0 | 0 | $\bar{A} \bar{B} \bar{C} = m_0$ | $A + B + C = M_0$                   |
| 0         | 0 | 1 | $\bar{A} \bar{B} C = m_1$       | $A + B + \bar{C} = M_1$             |
| 0         | 1 | 0 | $\bar{A} B \bar{C} = m_2$       | $A + \bar{B} + C = M_2$             |
| 0         | 1 | 1 | $\bar{A} B C = m_3$             | $A + \bar{B} + \bar{C} = M_3$       |
| 1         | 0 | 0 | $A \bar{B} \bar{C} = m_4$       | $\bar{A} + B + C = M_4$             |
| 1         | 0 | 1 | $A \bar{B} C = m_5$             | $\bar{A} + B + \bar{C} = M_5$       |
| 1         | 1 | 0 | $A B \bar{C} = m_6$             | $\bar{A} + \bar{B} + C = M_6$       |
| 1         | 1 | 1 | $A B C = m_7$                   | $\bar{A} + \bar{B} + \bar{C} = M_7$ |

Table 4.7 Minterms and Maxterms for three variables

Figure 31: Minterms and maxterms for three variables.

**Q12. Explain complementing Boolean functions and algebraic manipulation of expressions with examples.**

**Answer:**

**Complementing a function:** interchange the AND and OR operators and complement every literal (De Morgan's theorems).

**Example 1:**  $f = x'yz' + xy'z' \Rightarrow f' = (x + y' + z)(x' + y + z)$

**Example 2:**  $g = (a' + bc)d' + e \Rightarrow g' = (a(b' + c') + d) e'$

**Algebraic manipulation:** Boolean expressions are simplified using the laws of Boolean algebra, which reduces the number of gates required. The worked example below shows a step-by-step simplification.

- Interchange AND and OR operators
- So, what is the complement of  $f = x'yz' + xy'z'$ ?

**Answer:**  $f' = (x + y' + z)(x' + y + z)$

- **Example 2:** Complement  $g = (a' + bc)d' + e$

**Answer:**  $g' = (a(b' + c') + d)e'$

Figure 32: Complementing Boolean functions — worked answers.

- The objective is to acquire skills in manipulating Boolean expressions, to transform them into simpler form.
- **Example 1:** prove  $x + xy = x$  (absorption theorem)
- **Proof:**

$$\begin{aligned} x + xy &= x \cdot 1 + xy & x \cdot 1 &= x \\ &= x \cdot (1 + y) & \text{Distributive } \cdot \text{ over } + \\ &= x \cdot 1 = x & (1 + y) &= 1 \end{aligned}$$
- **Example 2:** prove  $x + x'y = x + y$  (simplification theorem)
- **Proof:**

$$\begin{aligned} x + x'y &= (x + x')(x + y) & \text{Distributive } + \text{ over } \cdot \\ &= 1 \cdot (x + y) & (x + x') &= 1 \\ &= x + y \end{aligned}$$

Figure 33: Algebraic manipulation — worked example.